JIHYUNG KOOK

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RESEARCH INTERESTS

Verifiable and Privacy-Preserving Computation; Applied Cryptography (Zero-Knowledge Proofs, Homomorphic Encryption, Multi-Party Computation); AI Security & Privacy

EDUCATION

Georgia Institute of Technology **Atlanta, GA**
Online M.S. in Computer Science Dec. 2024

- Specialization: Computational Perception & Robotics

Sookmyung Women's University (SMWU) **Seoul, Korea**
B.S. in Statistics and Computer Science Feb. 2017

MANUSCRIPTS

[1] **Kook, J.**, Kang, D., Lee, E. "Sparsification-Friendly CKKS-Based Federated Learning via RLWE-to-MLWE Conversion," *Manuscript in preparation.*

[2] Yoo, S., Oh, T., **Kook, J.**, et al., "Association between Muscle Cramping and Diabetic Peripheral Neuropathy: Findings from a Prospective Cohort Study," *Diabetes & Metabolism Journal.* Under review, Jun. 2026.

PUBLICATIONS & PRESENTATIONS

[1] **Kook, J.**, Kim, S., and Vakudavathu, P. C. S., "Sensitive Information Leakage from Tool-Call Traces," *Poster accepted at the AIWild and InfPriv Workshops in NeurIPS,* Dec. 2026 (https://neurips.cc/virtual/2026/loc/sydney/168249).

[2] **Kook, J.** and Lee, E., "Efficiency Evaluation of Sparsified Homomorphic Encryption in Federated Learning," *Poster presented at the Korean Institute of Communications and Information Sciences (KICS) Fall Conference*, pp. 752–753, Nov. 2025 (https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE12567025).

[3] Oh, T., **Kook, J.**, Jung, S., et al., "A Standardized Glucose-Insulin-Potassium Infusion Protocol in Surgical Patients: Use of Real Clinical Data from a Clinical Data Warehouse," *Diabetes Research and Clinical Practice*, vol. 174, article 108756, Apr. 2021 (DOI: 10.1016/j.diabres.2021.108756).

[4] Yoon, H., **Kook, J.**, Shim, J., "IoT-based Mailbox System Using Android and Arduino," *Poster presented at the Korea Information Processing Society (KIPS) Spring Conference*, Apr. 2015 (DOI: 10.3745/PKIPS.y2015m04a.1080).

RESEARCH EXPERIENCE

Independent Research—Adversarial Analysis of Verifiable Model Certification **Remote**
Co-organizer, Georgia Tech OMSCS ZKP Study Group Aug. 2026 – Present

- Analyzed fixed-point tolerance constraints in FairProof's Go/gnark implementation; adapted its existing circuits to introduce a malicious-prover mode that submits inflated distance witnesses while satisfying the existing constraints.
- Constructed a synthetic ReLU-network proof of concept that inflated the reported fairness radius from 0.0277 to 0.8775 (31.6×), with all 24 logged component checks passing under locally configured tolerances.
- Reproduced collaborator-provided output-layer scaling experiments on the supplied model, observing up to 14.7% radius inflation relative to honest runs with all 776 Groth16 verification checks passed.
- Prepare extension of testing to deeper networks with more parameters and study FairZK.

Sejong University—Privacy-Preserving Machine Learning and Cryptography Lab **Seoul, Korea**
Full-time Researcher (Advisor: Prof. Eunsang Lee) Jul. 2025 – Present

- Led experimental design for threshold-CKKS federated learning, screening chunk sizes and selection ratios in plaintext before evaluating CIFAR-10/100 configurations using OpenFHE/C++.
- Demonstrated the benefit of finer-grained update selection: reducing chunk size from 8,192 to 128 at ρ=0.2 improved peak accuracy by 6.46 and 6.73 percentage points on CIFAR-10 and CIFAR-100 respectively.
- Reduced cumulative online communication by 78.8-80.4% via sparse aggregation (ρ=0.2, B=128), with peak accuracy losses of 1.59 and 3.42 percentage points on CIFAR-10/100, respectively, versus full encrypted aggregation.

Independent Research—Privacy Leakage in Agentic AI **Remote**
Research Collaborator Jul. – Sep. 2026

- Developed and evaluated three- and four-layer MLP classifiers on 3,190 simulated healthcare-agent traces to infer sensitive patient attributes from tool names alone, without access to prompts, tool arguments, or outputs.
- Compared order-agnostic bag-of-tools frequency features with chronological n-gram features across three sensitive-attribute prediction tasks; the model achieved a macro-F1 of 0.689 for treatment complexity, demonstrating privacy leakage even without invocation-order information.

Seoul National University Bundang Hospital (SNUBH) **Seongnam, Korea**
Part-time Researcher, Endocrinology & Metabolism (Advisor: Prof. Tae Jung Oh) Mar. 2018 – Jul. 2020

- Conducted data management for a retrospective study evaluating a standardized glucose-insulin-potassium infusion protocol using records from SNUBH's Clinical Data Warehouse.
- Performed statistical analyses in R for 1:1 propensity score matching (227 patients per group), linear interpolation of blood glucose measurements, and comparisons of glucose AUC and time within the target glucose range.
- Engineered Python features from continuous glucose monitoring (CGM) data of 55 patients, achieving an AUC of up to 0.88 with support vector classification using peak-timing and glucose-variability measures.

PROFESSIONAL EXPERIENCE

Seoul National University Bundang Hospital **Seongnam, Korea**
Part-time Research Assistant, Endocrinology & Metabolism Jul. 2025 – Present

- Preprocess, merge, and analyze EMR datasets in Python, generating descriptive and sex-stratified comparison tables and univariable logistic regression results with odds ratios and 95% confidence intervals for clinical manuscripts.
- Coordinate administration and support site preparation for a Korea Disease Control and Prevention Agency-funded multicenter pragmatic randomized trial of personalized risk score-based management for Type II diabetes elderly.

KB Kookmin Bank & KB Financial Group **Seoul, Korea**
Data Analyst, Department of Data Planning Jul. 2020 – Sep. 2023

- Standardized customer data across 7 subsidiaries into 8 domains and implemented retention/deletion rules for a group-wide database.
- Coordinated quality reviews supporting cleanup of 20% legacy records and updates to 10% of high-value datasets.
- Managed scenario- and product-specific customer journey mapping across 7 subsidiaries to identify drop-off.
- Delivered SQL, Tableau, and automation training and individual coaching across 7 subsidiaries, updating 30+ topics annually to enable self-service analytics.

Croquis Inc. (now Kakao Style) **Seoul, Korea**
Junior Data Analyst, Department of Data Mar. 2018 – Apr. 2020

- Automated 17 ETL workflows using Python, PySpark, and SQL to standardize customer analysis, improving data accuracy and reliability by 25%.
- Designed and evaluated A/B tests for a new-arrivals feed to inform rollout; developed a seller clustering model using categorical and numerical attributes to benchmark merchant performance.
- Defined in-app payment KPIs targeting the top 100 sellers among 400+ shops; built Tableau/R dashboards and automated promotion reports. Early-adopting sellers recorded monthly sales 1.5× their pre-adoption levels.
- Delivered SQL and Tableau training to expand self-service analytics, reducing the data team's workload by 10%.

ADDITIONAL TRAINING

Seoul National University Big Data Institute **Seoul, Korea**
4th Industrial Revolution Academy: Big Data Platform Technology Jul. 2017 – Feb. 2018

- Completed intensive training in Python, SQL/DBMS, distributed systems, data mining, and machine/deep learning.
- Conducted an exploratory study at SNUBH to investigate predictive factors in CGM data under faculty supervision.

PROFICIENCY IN SKILLS

Programming: Python, C/C++, Go, SQL, R, PyTorch, PySpark
Cryptography, Privacy & Systems: Homomorphic Encryption (CKKS), OpenFHE, Federated Learning, Zero-Knowledge Proofs (gnark), Git, Docker, Linux
Data Analysis & Research Tools: Statistical Modeling, Matplotlib, ggplot2, Shiny, Tableau, LaTeX